

First, I need to determine the number of ways to draw exactly two hearts from the deck.

["end-section"]

["adding-knowledge"]

There are 13 hearts in the deck, and I want to choose 2 of them.

["end-section"]

["deduction"]

The number of ways to do this is 13 choose 2.

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["adding-knowledge"]

Then, I need to choose 1 card from the remaining 27 non-heart cards.

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["deduction"]

The number of ways to do this is 27 choose 1.

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["deduction"]

Finally, I multiply these two numbers together to get the total number of favorable outcomes.

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["deduction"]

Then, I divide by the total number of possible ways to draw 3 cards from the deck, which is 52 choose 3.

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["deduction"]

This gives me the probability of drawing exactly two hearts.

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